

# Finite Group Theory

Lectured by Kalyani Kansal

Scribed by Yu Coughlin

Autumn 2025

## Contents

|          |                                           |          |
|----------|-------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                       | <b>1</b> |
| 1.1      | Basic constructions                       | 1        |
| 1.2      | Actions                                   | 2        |
| <b>2</b> | <b>Composition series</b>                 | <b>2</b> |
| <b>3</b> | <b>Finite simple groups</b>               | <b>2</b> |
| 3.1      | Alternating groups                        | 2        |
| 3.2      | Matrix groups                             | 2        |
| 3.3      | Iwasawa's lemma                           | 2        |
| <b>4</b> | <b>Sylow's theorems</b>                   | <b>2</b> |
| <b>5</b> | <b>Extensions and semidirect products</b> | <b>2</b> |
| 5.1      | Groups of order $pq$                      | 2        |
| <b>6</b> | <b>Solvable groups</b>                    | <b>2</b> |
| 6.1      | Hall's theorem                            | 2        |
| <b>7</b> | <b>Nilpotent groups</b>                   | <b>2</b> |
| 7.1      | Solvable radical and fitting subgroup     | 4        |
| <b>8</b> | <b>The Frattini subgroup</b>              | <b>5</b> |
| 8.1      | Groups of order $p^3$                     | 5        |
| 8.2      | Groups of order 16                        | 6        |
| <b>9</b> | <b>The transfer homomorphism</b>          | <b>6</b> |

## 1 Introduction

### 1.1 Basic constructions

A recap of some basic notions we will assume familiarity with going forward.

**Proposition 1.1.1.** Let  $\phi : G \rightarrow H$  be a group homomorphism, then there is a canonical isomorphism:

$$G/\ker(\phi) \cong \text{im}(\phi)$$

**Proposition 1.1.2.** Let  $N \trianglelefteq G$ , then every subgroup of  $G/N$  is of the form  $G/H$  for some  $N \leq H \leq G$ .

**Definition 1.1.3.** A subgroup  $H \leq G$  is *characteristic* if for all  $\phi \in \text{Aut}(G)$ ,  $\phi(H) = H$ .

This is a stronger condition than being a normal subgroup which only requires invariance under inner automorphisms. If  $G \cong C_2 \times C_2$ , then  $G$  has no characteristic subgroups, as there is an automorphism swapping the two  $C_2$  factors, so this is a strictly stronger condition than just being normal.

**Proposition 1.1.4.** If  $M, N \leq G$  and  $M \text{ char } N$ ,  $N \trianglelefteq G$ , then  $M \trianglelefteq G$ .

*Proof.* If  $M$  is just normal in  $G$  then we don't know it to be invariant under all conjugations, just those from  $N$ , but now  $G$  is closed under all automorphisms of  $N$ , this includes conjugation from elements of  $G$ , so  $M$  is normal.  $\square$

**Proposition 1.1.5.** The commutator  $G' := [G, G]$  is a characteristic subgroup of  $G$ , and  $G/G'$  is abelian. For any normal subgroup  $N \trianglelefteq G$ ,  $G/N$  is abelian iff  $G' \leq N$

*Proof.*  $\square$

**Proposition 1.1.6.** If  $A, B \trianglelefteq G$ ,  $A \cap B = \{1\}$ , and  $G = AB$ , then  $G \cong AB$ .

**Theorem 1.1.7.** If  $G$  is a finitely generated abelian group, then there exists a positive integer  $m \geq 0$  and prime powers  $p_1^{\alpha_1}, \dots, p_n^{\alpha_n}$  such that:

$$G \cong \mathbb{Z}^m \times C_{p_1^{\alpha_1}} \times \cdots \times C_{p_n^{\alpha_n}}$$

said  $m$  and the  $p_i^{\alpha_i}$  are uniquely determined (up to reordering of the prime powers) by  $G$ .

## 1.2 Actions

**Proposition 1.2.1.** If  $G \curvearrowright \Omega$ , then for any  $\alpha \in \Omega$ , the size of the orbit  $\alpha^G$  equals  $|G : G_\alpha| = |G|/|G_\alpha|$  where  $G_\alpha \leq G$  is the stabiliser of  $\alpha$  in  $G$ .

**Corollary 1.2.2** (Cauchy's theorem). If  $G$  is a finite group whose order is divisible by a prime  $p$ , then  $G$  contains an element of order  $p$

## 2 Composition series

## 3 Finite simple groups

### 3.1 Alternating groups

### 3.2 Matrix groups

### 3.3 Iwasawa's lemma

## 4 Sylow's theorems

## 5 Extensions and semidirect products

### 5.1 Groups of order $pq$

## 6 Solvable groups

**Proposition 6.0.1** (Frattini Argument). Let  $N \trianglelefteq G$  and  $P \in \text{Syl}_p(N)$ , then  $G = N_G(P)N$ .

*Proof.* For all  $g \in G$ , we have  ${}^gP \leq {}^gN$ , thus  ${}^gP \in \text{Syl}_p(N)$  and by Sylow 4, there exists some  $n \in N$  with  ${}^gP = {}^nP$ , thus  $gn^{-1} \in N_G(P)$ , and  $g \in N_G(P)N$ .  $\square$

### 6.1 Hall's theorem

## 7 Nilpotent groups

**Definition 7.0.1.** A finite group  $G$  is **nilpotent** if there exists a series of subgroups:

$$1 = G_0 \leq G_1 \leq \cdots \leq G_r = G$$

such that for all  $0 \leq i \leq r-1$

$$G_i \trianglelefteq G \text{ and } G_{i+1}/G_i \leq Z(G/G_i)$$

such a series of subgroups is called a **central series**

Notably, all abelian groups are nilpotent; and all nilpotent groups are solvable, as  $G_{i+1}/G_i \leq Z(G/G_i)$  implies  $G_{i+1}/G_i$  is abelian.

**Proposition 7.0.2.** Every finite  $p$ -group is nilpotent.

*Proof.* Let  $P$  be such a nontrivial  $p$ -group, define a series of subgroups as follows:

- $P_0 := 1$ ;
- $P_1 := Z(P)$ ;
- Have  $P_2$  a subgroup containing  $P_1$  such that  $P_2/P_1 = Z(P/P_1)$ ;
- recursively define each  $P_{i+1}$  to be a subgroup containing  $P_i$  such that  $P_{i+1}/P_i = Z(P/P_i)$ .

as nontrivial  $p$ -groups always have nontrivial centre this series is strictly increase, so must terminate at some  $P_r = P$ .  $\square$

**Definition 7.0.3.** The **upper central series** of a finite group  $G$  is the central series  $Z_i(G)$  formed in the above proposition. We are taking the *biggest step* possible at each point.

**Definition 7.0.4.** For  $G$  a finite group and  $H, K \subseteq G$ , define

$$[H, K] := \langle [h, k] \mid h \in H, k \in K \rangle \leq G$$

one can immediately see  $[H, K] = [K, H]$ . We want to define the central series that takes the biggest step down at each point: this will be the **lower central series** of  $G$  defined as follows:

- $\gamma_0(G) := G$ ;
- For  $\gamma_1(G)$  to be the next step down it must satisfy

$$G/\gamma_1(G) \leq Z(G/\gamma_1(G))$$

which just means  $G/\gamma_1(G)$  is abelian, so we must have  $[G, G] \leq \gamma_1(G)$ , if we want to make the biggest just down  $\gamma_1(G)$  must be as small as possible so we set  $\gamma_1(G) := [G, G]$ ;

- Now given our  $\gamma_i(G)$ , to construct  $\gamma_{i+1}(G)$  we must satisfy:

$$\gamma_i(G)/\gamma_{i+1}(G) \leq Z(G/\gamma_{i+1}(G))$$

so  $\gamma_i(G)$  should commute with everything in  $G$  up to  $\gamma_{i+1}(G)$  which is just:

$$[G, \gamma_i(G)] \leq \gamma_{i+1}(G)$$

so to continue with our biggest steps, we should set  $\gamma_{i+1}(G) := [\gamma_i(G), G]$ .

We should expect our two intuitively shortest possible sequences to be of the same minimal length, i.e. our group is nilpotent iff these series terminate. This is true, we will first need a lemma to prove it.

**Lemma 7.0.5.** For  $G$  a finite group and  $X, Y \leq G$ :

1.  $Y \leq N_G(X)$  iff  $[X, Y] \leq X$ ;
2. If  $Y \trianglelefteq G$  and  $Y \leq X$  then we have

$$X/Y \leq Z(G/Y) \iff [X, G] \leq Y$$

*Proof.* (1)  $Y \leq N_G(X)$  iff for all  $x \in X$  and  $y \in Y$  we have  $[x, y] = xyx^{-1}y^{-1} \in X$  iff  $[X, Y] \leq X$ .

(2) For all  $x \in X$  we have:  $xY \in Z(G/Y)$  iff  $[xY, gY] = Y$  for all  $g \in G$  iff  $[x, g]Y = Y$  iff  $[x, g] \in Y$ .  $\square$

This formalises our intuition of taking the biggest steps possible in the definitions of our upper and lower central series.

**Theorem 7.0.6.** For  $G$  a finite group and  $n$  a positive integer,  $Z_n(G) = G$  iff  $\gamma_n(G) = 1$ . Furthermore,  $G$  is nilpotent iff these conditions hold for some  $n$ .

*Proof.* If either of the statements are true, then  $G$  is certainly nilpotent. So it suffices to prove for nilpotent  $G$  the set of  $n$  such that  $Z_n(G) = G$  and  $\gamma_n(G) = 1$  are equal and nonempty.

Let

$$1 = N_0 \leq N_1 \leq \cdots \leq N_k = G$$

be a central series for  $G$ . By the previous lemma, for each  $i$  we have  $[N_{i+1}, G] \leq N_i$ .

Consider  $\gamma_0(G) = G \leq N_k$ , so  $\gamma_1(G) = [G, G] \leq [N_k, G] \leq N_{k-1}$  and inductively:

$$\gamma_i(G) = [\gamma_{i-1}(G), G] \leq [N_{k-i+1}, G] \leq N_{k-i}$$

and  $N_0 = 1$  we must have  $\gamma_k(G) = 1$ , applying this to our upper central series is half way.

Now consider  $Z_0(G) = 1 = N_0(G)$ , assume for  $i \geq 1$  that  $N_{i-1} \leq Z_{i-1}(G)$ , equivalently:

$$[N_i, G] \leq N_{i-1} \leq Z_{i-1}(G)$$

moreover  $[N_i Z_{i-1}(G), G] \leq Z_{i-1}$  as for all  $n \in N_i$ ,  $z \in Z_{i-1}(G)$ , and  $g \in G$  we have

$$[nz, g] Z_{i-1}(G) = [n, zg][z, g] Z_{i-1}(G) \subseteq [N_i, G] Z_{i-1}(G) \leq Z_{i-1}(G)$$

hence

$$N_i Z_{i-1}(G) / Z_{i-1}(G) \leq Z(G / Z_{i-1}(G)) = Z_i(G) / Z_{i-1}(G)$$

so  $N_i \leq Z_i(G)$  and by induction we are done.  $\square$

**Definition 7.0.7.** The a nilpotent group  $G$  the smallest integer  $n$  such that  $Z_n(G) = G$  (or equivalently  $\gamma_n(G) = 1$ ) is called the nilpotency class of  $G$ .

**Corollary 7.0.8.** Subgroups and quotients of nilpotent groups are also nilpotent.

*Proof.* Let  $G$  have nilpotency class  $n$ , if  $H \leq G$  then  $\gamma_n(H) \leq \gamma_n(G) = 1$  so  $H$  is nilpotent.

Let  $\phi$  be the quotient map  $G \rightarrow G/N$  for some  $N \trianglelefteq G$ , as homomorphisms commute with commutators and  $\phi$  is surjective, we have  $\phi(\gamma_i(G)) = \gamma_i(\phi(G))$  for all  $i$ , specifically  $\gamma_n(\phi(G)) = 1$  so  $G/N$  is nilpotent.  $\square$

If such an  $N$  and  $G/N$  are both nilpotent,  $G$  is not necessarily nilpotent. For example take  $S_3 = C_3 \rtimes C_2$  which isn't nilpotent:  $[S_3, S_3] = C_3$ ,  $[C_3, S_3] = C_3$ .

Nilpotent groups have lots of very nice characterisations.

**Theorem 7.0.9.** For a finite group  $G$  the following are equivalent:

1.  $G$  is nilpotent;
2. for any  $H < G$ , we have  $H < N_G(H)$  is proper;
3. every maximal subgroup of  $G$  is normal;
4. every Sylow subgroup of  $G$  is normal;
5.  $G \cong P_1 \times \cdots \times P_k$  for  $P_i$  a  $p_i$ -group with all the  $p_1, \dots, p_k$  distinct primes.

*Proof.* (1  $\Rightarrow$  2) Let  $1 = N_0 \leq \cdots \leq N_k = G$  be a central series, and let  $H < G$ . Let  $r$  be the largest positive integer such that  $N_r \leq H$ , then  $N_{r+1}/N_r \leq Z(G/N_r)$  and equivalently  $[N_{r+1}, H] \leq N_r \leq H$ , by our lemma we know this is equivalent to  $N_{r+1} \leq N_G(H)$ , hence  $H \leq N_G(H)$ .

(2  $\Rightarrow$  3) Let  $M$  be a maximal subgroup of  $G$ , then  $M \leq N_G(M) = G$ , as  $M$  is maximal.

(3  $\Rightarrow$  4) Let  $P \in \text{Syl}_p(G)$  with  $P$  not normal in  $G$ , so  $N_G(P) < G$ . There exists a maximal subgroup  $M$  of  $G$  containing  $N_G(P)$  with  $M \triangleleft G$ . As  $P \in \text{Syl}_p(M)$  we can use the Frattini argument to get  $G = N_G(P)M = M$ , a contradiction, thus  $P \trianglelefteq G$ .

(4  $\Rightarrow$  5) Let  $P_1, \dots, P_k$  be the Sylow subgroups of  $G$  for distinct primes, as each  $P_i$  is normal in  $G$ ,  $[P_i, P_j] \leq P_i \cap P_j = 1$ , hence  $P_1 P_2 \cdots P_k = G$  with each  $g$  written uniquely as a product of elements from different  $P_i$ .

(5  $\Rightarrow$  1) Products commute with the  $\gamma_i$ , so the finite maximum of all the components nilpotency classes will be the nilpotency class of  $G$ .  $\square$

## 7.1 Solvable radical and fitting subgroup

## 8 The Frattini subgroup

**Definition 8.0.1.** The  $G$  a finite group the intersection all all maximal subgroups is called the **Frattini subgroup** and denoted  $\Phi(G)$ .

**Proposition 8.0.2.** For all finite groups  $G$  and  $x \in G$  the following are equivalent:

1.  $x \in \Phi(G)$ ;
2. For all  $S \subseteq G$ , if  $G = \langle S, x \rangle$ , then  $G = \langle S \rangle$ .

*Proof.* (1  $\Rightarrow$  2) Suppose  $x \in \Phi(G)$  and there exists some  $S \subset G$  such that  $\langle S, x \rangle = G$  but  $\langle S \rangle \neq G$ , then there exists some maximal subgroup  $M < G$  containing  $S$ , but  $x \in \Phi(G) \leq M$ , thus  $G = \langle S, x \rangle \leq M$ , a contradiction.

(2  $\Rightarrow$  1) If  $x \in G$  satisfies property 2 and  $x \notin \Phi(G)$  then there exists some maximal  $M < G$  with  $x \notin M$ , thus  $\langle M, x \rangle = G$ , but  $\langle M \rangle \neq G$ , a contradiction.  $\square$

We call elements of  $\Phi(G)$  the **non-generators** of  $G$ .

**Proposition 8.0.3.**  $\Phi(G)$  is a characteristic nilpotent subgroup of  $G$ .

*Proof.* An automorphism of  $G$  must permute the maximal subgroups, so fixes their intersection, thus  $\Phi(G)$  char  $G$ .

Let  $p$  be a prime and  $P \in \text{Syl}_p(\Phi(G))$ , by the Frattini argument, we must have  $G = N_G(P)\Phi(G)$ . So by the previous proposition (and some induction)  $G = N_G(P)$  so  $P \trianglelefteq G$ , and a fortiori  $G \trianglelefteq \Phi(G)$ . Now, by the characterisation of nilpotent subgroups,  $\Phi(G)$  is nilpotent.  $\square$

**Theorem 8.0.4** (Burnside Basis Theorem). For  $P$  a  $p$ -group with  $|P| = p^n$  and  $|\Phi(P)| = p^{n-d}$  for some  $d \geq 1$ , we have the following:

1.  $P/\Phi(P) \cong \mathbb{F}_p^d$  the elementary abelian group of order  $p^d$ ;
2. If  $N \trianglelefteq P$  and  $P/N$  is an elementary abelian group, then  $\Phi(P) \leq N$ , thus  $\Phi(G)$  is the unique normal subgroups with minimal elementary abelian quotient;
3. For  $x_1, \dots, x_d \in P$  which equivalence classes  $\overline{x_i} \in P/\Phi(P)$  respectively:

$$P = \langle x_1, \dots, x_d \rangle \iff \overline{x_1}, \dots, \overline{x_d} \text{ is a basis for } \mathbb{F}_p^d$$

*Proof.* (1) We know any maximal subgroup  $M < P$  is normal, for a maximal normal subgroup the quotient is simple, so  $|P/M| = p$ . Thus  $P' \leq M$  and  $x^p \in M$  for all  $x \in P$ , so  $P' \leq \Phi(P)$  and  $P/\Phi(P)$  is elementary abelian.

(2) Suppose  $N \trianglelefteq P$  and  $P/N$  is elementary abelian, then  $\Phi(P/N) = 1$ . The maximal subgroups of  $P/N$  are of the form  $M/N$  for some maximal  $M < P$  containing  $N$ . As  $\Phi(P/N) = N$ , the intersection of these must be contained in  $N$ , thus  $\Phi(P) \leq N$ .

(3) If  $x_1, \dots, x_d$  span  $P$ , then they must span  $\mathbb{F}_p^d$  and so form a basis. If  $\overline{x_1}, \dots, \overline{x_d}$  is a basis for  $\mathbb{F}_p^d$  then  $P = \langle x_1, \dots, x_d, \Phi(P) \rangle$  and thus  $P = \langle x_1, \dots, x_d \rangle$ .  $\square$

We can use this to relate the minimal number of generators of  $P$  to its Frattini subgroup:

The minimal number of generators of  $D_8$  is 2, so  $D_8/\Phi(D_8) \cong C_2^2$ , the only order two subgroup  $\Phi(D_8)$  can be is  $\langle r^2 \rangle$ .

### 8.1 Groups of order $p^3$

The only ones left are non-abelian groups of order  $p^3$  with  $p$  odd.

**Lemma 8.1.1.** Let  $G$  be a group such that  $G' \leq Z(G)$ , then for all  $x, y \in G$  and  $n \geq 1$ :

$$x^n y^n = (xy)^n [x^{-1}, y^{-1}]^{n(n-1)/2}$$

*Proof.* We go by induction on  $n$ , noting:

$$xy = yx[x^{-1}, y^{-1}]$$

and that by assumption  $[x^{-1}, y^{-1}] \in Z(G)$ . Assume the claim is true for  $n$  and consider:

$$\begin{aligned} (xy)^{n+1}[x^{-1}, y^{-1}]^{n(n+1)/2} &= (xy)^n[x^{-1}, y^{-1}]^{n(n-1)/2}xy[x^{-1}, y^{-1}]^n \\ &= x^ny^n x[x^{-1}, y^{-1}]^n y = x^ny^{n-1}xy[x^{-1}, y^{-1}]^{n-1}y \\ &\vdots \\ &= x^{n+1}y^{n+1} \end{aligned}$$

□

**Theorem 8.1.2.** For  $p$  an odd prime, there are two non-abelian groups of order  $p^3$  up to isomorphism:

1.  $P_1 = \langle a, b, c \mid a^p = b^p = c^p = 1, [a, b] = c, [a, c] = [b, c] = 1 \rangle \cong \langle a, c \rangle \rtimes \langle b \rangle \cong C_p^2 \rtimes C_p$ ;
2.  $P_2 = \langle a, b \mid a^{p^2} = b^p = 1, [a, b] = a^p \rangle \cong \langle a \rangle \rtimes \langle b \rangle \cong C_{p^2} \rtimes C_p$ .

*Proof.* Let  $P$  be such a group, then By the Burnside basis theorem  $P/\Phi(P) \cong C_p^d$  for one of  $d = 1, 2, 3$ . If  $d = 1$ , then  $P$  is cyclic; if  $d = 3$  then  $P \cong C_p^3$  abelian, so we must have  $d = 2$ . Write:

$$P = \langle a, b \rangle, \quad \Phi(P) = \langle c \rangle$$

for some  $a, b, c \in P$ . We also know  $\Phi(P) = Z(P) = P'$ .

First, assume  $a^p = b^p = 1$ . As  $[a, b] \neq 1$  ( $P$  non-abelian), take  $c = [a, b]$ , this gives all the relations for  $P_1$ , so this case is done.

Secondly, assume  $a$  has order  $p^2$ , from the proof of the Burnside basis theorem we know  $a^p \in \Phi(P)$ , so  $\Phi(P) = \langle a^p \rangle$ . If  $b^p = 1$ , then by taking  $[a, b] = a^p$  we have all the relations for  $P_2$ .

If, instead, we let  $b$  have order  $p^2$ , wlog we may assume  $a^p = c$  and  $b^p = c^{-1}$  (by choosing different powers of  $a$  and  $b$ ). Now by the previous lemma we can write

$$1 = a^p b^p = (ab)^p [a^{-1}, b^{-1}]^{p(p-1)/2}$$

where  $p$  being odd must divide  $p(p-1)/2$ , thus  $[a^{-1}, b^{-1}]^{p(p-1)/2} = 1$  and hence  $(ab)^p = 1$  and by replacing  $b$  with  $ab$  we reach the first state. □

$P_1$  also has an upper unitriangular representation over  $\mathbb{F}_p^3$  given by:

$$a \mapsto \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad b \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad c \mapsto \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

## 8.2 Groups of order 16

Not lectured.

**Definition 8.2.1.** Here are four new classes of groups of order  $p^n$  for  $n \geq 3$ :

- $\text{Mod}_{p^n}$
- $D_{2^n}$
- $SD_{2^n}$
- $Q_{2^n}$

## 9 The transfer homomorphism